CHM-DSM-401 — Fundamental of Chemistry-IV

Unit V: Elementary Food Chemistry

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: CHM-DSM-401, Full Marks 100 [ESE 70 + Internal 30] | Pass marks 40 | Unit V: 9 lectures

Notes 20 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

1. Food additives: vitamins, amino acids, minerals.
2. Aroma compounds as food flavours.
3. Sugar substitutes: sorbitol, saccharin, cyclamate.
4. Food colours: types and compositions.
5. Proteins and meat, carbohydrates, lipids and emulsions.
6. Browning reactions.
7. Preservation of foods: general principles, physical and chemical methods.

Top 2 high-mark topics in this unit

1. Preservation of foods - general principles, physical and chemical methods of food preservation — the syllabus closes the unit with this, a classic long descriptive answer with methods and examples.
2. Food additives - vitamins, amino acids, minerals, aroma compounds, sugar substitutes and food colours — a list-heavy topic, and AU Biomolecules asks vitamins, amino acids and lipids in the same short/long style.

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1. Food Additives: Vitamins, Amino Acids and Minerals

Definition and Purpose of Food Additives

Food additives are chemical substances intentionally added to food products during processing, storage, packaging, or transportation to achieve specific technological functions. The primary objectives of incorporating food additives into food systems include improving keeping quality by inhibiting chemical deterioration or microbial spoilage, enhancing organoleptic attributes such as flavour, appearance, and texture, and preserving or augmenting nutritive value.

Food additives are categorized based on their functional role within a food matrix. Preservatives extend shelf-life by preventing oxidation or microbial proliferation, texturizing agents modify viscosity and phase stability, and nutritional fortificants restore or enrich the essential nutrient content of processed foods. Because food additives are consumed regularly, strict regulatory control is required to establish permitted limits. Legally defined limits prevent acute and chronic toxicity, ensure that additives do not mask food spoilage or deceive consumers, and guarantee that the cumulative dietary intake remains within safe physiological thresholds.

Vitamins as Food Additives

Vitamins are organic micro-constituents required in trace amounts for fundamental metabolic processes, cellular homeostasis, and enzymatic reactions. They are classified based on their solubility into fat-soluble vitamins and water-soluble vitamins. Both classes are widely used as food additives to restore nutrient losses incurred during processing or to fortify foods lacking these essential compounds.

Fat-Soluble Vitamins

Fat-soluble vitamins—comprising vitamins A, D, E, and K—are hydrophobic molecules that are absorbed and transported alongside dietary lipids.

Vitamin A (retinol and its provitamin precursors such as β-carotene) plays a critical role in the visual cycle by forming the chromophore rhodopsin within rod photoreceptor cells. It also regulates epithelial tissue integrity and cellular differentiation. In food processing, vitamin A is frequently added to margarine, edible oils, and skimmed milk to compensate for fat removal or to match the nutritional profile of natural butter.

Vitamin D exists primarily as ergocalciferol (vitamin D_2) and cholecalciferol (vitamin D_3). It functions as a precursor to the hormone calcitriol ($1,25\text{-dihydroxyvitamin D}_3$), which maintains plasma calcium homeostasis by regulating intestinal calcium absorption and bone mineralization. Vitamin D is commonly added to fluid milk, plant-based milk substitutes, and breakfast cereals to prevent deficiency disorders such as rickets and osteomalacia.

Vitamin E encompasses a group of tocopherols and tocotrienols, with α-tocopherol demonstrating the highest biological activity. Structurally characterized by a chromanol ring with a lipophilic side chain, vitamin E serves as a potent lipid-soluble antioxidant. It quenches peroxyl radicals, thereby terminating lipid peroxidation chain reactions in unsaturated fatty acids:

$$\mathrm{ROO}^{\bullet} + \text{Toc-OH} \rightarrow \mathrm{ROOH} + \text{Toc-O}^{\bullet}$$

Vitamin E is added to vegetable oils, lard, fried snack foods, and infant formulas both as a nutrient and as a functional antioxidant to prevent rancidity.

Vitamin K (phylloquinone, vitamin K_1; menaquinones, vitamin K_2) acts as an essential cofactor for the γ-glutamyl carboxylase enzyme, which converts specific glutamic acid residues to γ-carboxyglutamate in proteins involved in the blood coagulation cascade, such as prothrombin. Vitamin K is routinely added to specialized infant formulas and medical nutrition preparations to ensure adequate clotting capability in neonates.

Water-Soluble Vitamins

Water-soluble vitamins encompass the B-complex vitamins and vitamin C. They generally act as coenzymes or antioxidant agents and are easily lost during aqueous food processing operations.

The vitamin B-complex includes thiamine (vitamin B_1), riboflavin (vitamin B_2), niacin (vitamin B_3), pyridoxine (vitamin B_6), folic acid (vitamin B_9), and cobalamin (vitamin B_{12}). Thiamine, riboflavin, and niacin serve as precursor molecules for key coenzymes—thiamine pyrophosphate (TPP), flavin adenine dinucleotide (FAD), and nicotinamide adenine dinucleotide (NAD^+)—which drive carbohydrate, lipid, and amino acid catabolism. Folic acid and cobalamin are required for single-carbon transfer reactions and nucleotide synthesis. B-complex vitamins are extensively added to refined cereal flours, polished rice, and breakfast cereals to replenish nutrients lost during milling.

Vitamin C (L-ascorbic acid) is an enediol compound that acts as a strong reducing agent and water-soluble antioxidant. Biologically, ascorbic acid acts as an essential cofactor for prolyl and lysyl hydroxylases, enzymes responsible for the post-translational modification of collagen. Ascorbic acid scavenges free radicals and protects cellular components from oxidative stress:

$$\text{Ascorbic Acid} + 2\mathrm{R}^\bullet \rightarrow \text{Dehydroascorbic Acid} + 2\mathrm{RH}$$

In food technology, ascorbic acid is added to fruit juices, soft drinks, cured meats, and dough systems. It functions as an antioxidant to prevent enzymatic browning, as a color stabilizer in cured meats, and as a dough conditioner in baking.

Amino Acids as Food Additives

Amino acids are the monomeric building blocks of proteins, characterized by a central α-carbon bonded to an amino group ($-\mathrm{NH}_2$), a carboxyl group ($-\mathrm{COOH}$), a hydrogen atom, and a variable side chain ($-\mathrm{R}$). In food chemistry, amino acids are classified into essential and non-essential amino acids based on human metabolic synthesis capacity.

Essential and Non-Essential Amino Acids

Essential amino acids cannot be synthesized *de novo* by the human body at a rate sufficient to meet physiological demands and must be obtained from dietary sources. An example of an essential amino acid is L-lysine. Lysine contains a basic ε-amino group on its side chain and serves as a limiting amino acid in cereal grains such as wheat, corn, and rice.

Non-essential amino acids can be synthesized endogenously by human metabolic pathways, typically via transamination reactions of intermediate metabolites. An example of a non-essential amino acid is L-glutamic acid. Glutamic acid contains an additional carboxylic acid group on its side chain and participates in amino acid metabolism and neurotransmission.

Use of Amino Acids in Food Enrichment

Many plant proteins exhibit incomplete amino acid profiles, lacking one or more essential amino acids. Food enrichment with free amino acids raises the overall biological value of food proteins. For instance, cereal products are fortified with L-lysine monohydrochloride to balance the amino acid profile of wheat protein (gliadin and glutenin), thereby improving the protein efficiency ratio (PER) of the final food product.

Amino acids are also added to enhance or modify food flavour. L-glutamic acid and its salt, monosodium glutamate (MSG), are added to savoury food products, soups, and snack foods to impart the *umami* basic taste modality. Furthermore, amino acids such as glycine and alanine contribute sweet flavour profiles, while specific combinations of amino acids undergo controlled Maillard reactions during baking to produce desirable aroma compounds.

Minerals as Food Additives

Minerals are inorganic elements essential for structural, physiological, and catalytic functions in biological systems. Major micronutrient minerals added to foods include calcium, iron, iodine, and zinc.

Calcium ($\mathrm{Ca^{2+}}$) is the primary structural mineral component of the human skeleton, occurring predominantly as hydroxyapatite, $\mathrm{Ca_{10}(PO_4)_6(OH)_2}$, in bones and teeth. It is also required for muscle contraction, neuronal signal transduction, and blood coagulation. Calcium salts such as calcium carbonate ($\mathrm{CaCO_3}$), calcium phosphate ($\mathrm{Ca_3(PO_4)_2}$), and calcium lactate are added to orange juice, plant-based beverages, bakery preparations, and breakfast cereals to augment daily dietary calcium intake and prevent bone density loss.

Iron (Fe) is a critical trace element required for oxygen transport and cellular respiration. It serves as the central coordination metal in the heme moiety of hemoglobin and myoglobin, and acts as an essential cofactor for cytochromes in the mitochondrial electron transport chain. Iron deficiency leads to nutritional anemia. Iron is added to wheat flour, cornmeal, rice, and infant formulas using fortificants such as ferrous sulfate ($\mathrm{FeSO_4}$), ferrous fumarate, and elemental iron powders.

Iodine (I) is essential for the biosynthesis of thyroid hormones, specifically thyroxine ($\mathrm{T_4}$) and triiodothyronine ($\mathrm{T_3}$), which regulate basal metabolic rate, protein synthesis, and cognitive development. Iodine deficiency results in endemic goiter and developmental impairments. Iodine is added to standard sodium chloride (NaCl) table salt in the form of potassium iodate ($\mathrm{KIO_3}$) or potassium iodide (KI), producing iodised salt.

Zinc ($\mathrm{Zn^{2+}}$) acts as a structural component and catalytic cofactor for over three hundred enzymes, including carbonic anhydrase, carboxypeptidases, and DNA/RNA polymerases. It is essential for immune function, protein synthesis, and wound healing. Zinc salts such as zinc oxide (ZnO) and zinc sulfate ($\mathrm{ZnSO_4}$) are added to fortified breakfast cereals, whole-grain flours, and infant formulas to combat dietary zinc deficiency.

Natural versus Synthetic Additives and Safety Limits

Food additives are classified according to their origin as either natural or synthetic additives. Natural additives are compounds isolated directly from biological matrices such as plants, animals, or microorganisms using physical or enzymatic extraction techniques. Examples include β-carotene extracted from carrots, lecithin extracted from soybeans, and ascorbic acid derived from citrus fruits. Synthetic additives are produced through chemical synthesis in industrial processes. Examples include synthetic ascorbic acid, potassium sorbate, and artificial colourants.

From a chemical perspective, a pure compound exhibits identical physical properties, chemical reactivity, and biological activity regardless of whether it was extracted from a natural source or synthesized in a laboratory. The primary distinction lies in potential impurities resulting from the extraction matrix or synthetic pathway reagents.

To protect public health, regulatory authorities establish safe intake levels for all approved food additives based on toxicological evaluation. The primary reference value derived from animal toxicological studies is the No Observed Adverse Effect Level (NOAEL), defined as the highest dose of a chemical substance that causes no detectable toxicological effect in test populations.

To account for inter-species extrapolation (differences between animals and humans) and intra-species variation (sensitivity differences among humans), a standard safety factor is applied. The standard safety factor value chosen by toxicologists is 100, which represents a 10-fold factor for inter-species variation multiplied by a 10-fold factor for intra-species variation:

$$\text{Safety Factor} = 10 \times 10 = 100$$

The Acceptable Daily Intake (ADI) is defined as the amount of a food additive, expressed on a body-weight basis, that can be consumed daily over a lifetime without appreciable health risk. The mathematical relationship defining the ADI from the NOAEL is:

$$\text{ADI} = \frac{\text{NOAEL}}{\text{Safety Factor}}$$

Substituting the standard safety factor value into this relationship yields:

$$\text{ADI} = \frac{\text{NOAEL}}{100}$$

The resulting ADI value, expressed in milligrams of additive per kilogram of body weight per day ($\text{mg/kg body weight/day}$), forms the scientific basis for determining the maximum allowable concentration limits of additives permitted in commercial food formulations.

2. Aroma Compounds as Food Flavours and Sugar Substitutes

Aroma Compounds as Food Flavours

Aroma compounds are volatile organic compounds (VOCs) that impart distinct smells and sensory properties to food materials. These compounds possess low molecular weights, typically less than $300\ \text{g/mol}$, and high vapour pressures at ambient temperature, allowing them to evaporate easily and reach the olfactory receptors located in the nasal cavity. The human perception of food flavour is a dual sensory mechanism combining taste, detected by taste buds on the tongue, and olfaction, detected via the retronasal and orthonasal pathways by olfactory sensory neurons. Volatile aroma compounds are primarily responsible for the characteristic odour profile and overall flavour sensation of fresh and processed foods.

Main Chemical Classes of Aroma Compounds

Aroma compounds belong to diverse chemical classes, categorized by their functional groups. The primary organic classes include esters, aldehydes, ketones, alcohols, and terpenes:

* **Esters:** Esters are synthesized chemically via the condensation reaction between a carboxylic acid and an alcohol. They are widely distributed in nature and generally impart sweet, fruity notes to fruits and flowers. A prominent example is ethyl acetate ($\text{CH}_3\text{COOCH}_2\text{CH}_3$), which provides a light, fruity aroma characteristic of apples, pears, and crushed fruits. * **Aldehydes:** Aldehydes contain a terminal carbonyl group ($-\text{CHO}$). They often possess intense aroma profiles with very low odor detection thresholds. * *Vanillin* (4-hydroxy-3-methoxybenzaldehyde) is the key flavour constituent of natural vanilla pods, producing a rich, sweet vanilla scent. * *Benzaldehyde* ($\text{C}_6\text{H}_5\text{CHO}$) is an aromatic aldehyde that exhibits a characteristic bitter almond odour, also present in cherry kernels and peach stones. * **Ketones:** Ketones feature an internal carbonyl group ($-\text{C}(=\text{O})-$) bonded to two carbon chains. They contribute to creamy, buttery, or fruity aroma profiles in dairy and fruit systems. * **Alcohols:** Organic compounds containing hydroxyl groups ($-\text{OH}$) act as important solvent bases and direct contributors to herbal, floral, or cooling sensory notes. * **Terpenes and Terpenoids:** Terpenes are natural hydrocarbons derived from repeating five-carbon isoprene units (C_5H_8). Oxygenated derivatives of terpenes are referred to as terpenoids. * *Citral* ($\text{C}_{10}\text{H}_{16}\text{O}$) represents an isomeric mixture of two terpenoid aldehydes (geranial and neral), imparting a fresh, strong lemon flavour found in citrus peels and lemongrass. * *Menthol* ($\text{C}_{10}\text{H}_{20}\text{O}$) is a monocyclic terpene alcohol extracted from mint oils, providing a distinct minty odour accompanied by a physical cooling sensation due to its interaction with thermal sensory receptors in the mouth.

Natural and Artificial Flavours

Aroma compounds utilized in food chemistry are broadly classified based on their origin:

* **Natural Flavours:** These are flavouring preparations extracted directly from natural plant or animal tissues using physical, microbiological, or enzymatic methods. Common sources include fruits, spices, herbs, roots, and essential oils. Natural flavours consist of complex mixtures containing dozens to hundreds of individual volatile organic components that collectively produce a natural flavour profile. * **Artificial Flavours:** These are flavouring substances manufactured entirely via chemical synthesis. They may either replicate the exact molecular structure of compounds found in nature (nature-identical flavourings) or feature synthetic chemical structures not identified in natural products but designed to evoke specific sensory responses. Artificial flavours are widely used due to their high chemical purity, thermal stability during food processing, lower production costs, and consistency between batches.

Industrial Applications in Foods and Drinks

Aroma compounds play a crucial role in the food and beverage industry:

1. **Restoration of Lost Aromas:** Processing methods such as pasteurization, drying, concentration, thermal sterilization, and baking often cause the evaporation or degradation of sensitive volatile compounds. Re-adding lost aroma compounds restores the original flavour profile of the processed product.
2. **Flavour Enhancement and Standardization:** In products like carbonated soft drinks, fruit juices, confectionery, ice creams, processed meats, and savory snacks, single or blended aroma compounds are incorporated to create uniform, intense, and appealing sensory characteristics.
3. **Masking Unpleasant Notes:** Aroma compounds are used to mask off-flavours arising from nutrient fortification, soy proteins, or metallic active ingredients in functional foods and dietary supplements.

Sugar Substitutes

Sugar substitutes are food additives that provide a sweet taste similar to that of sucrose (table sugar) while offering alternative metabolic profiles.

Rationale for Use

The primary reasons for replacing sucrose with sugar substitutes include:

* **Diabetes Management:** Sucrose is broken down into glucose and fructose, causing rapid elevation of blood glucose concentrations and demanding insulin release. Diabetics rely on non-nutritive or low-glycaemic sugar substitutes to enjoy sweet foods without inducing dangerous blood glucose spikes. * **Calorie Reduction and Weight Management:** High dietary intake of sucrose contributes to excess caloric consumption ($4.0\ \text{kcal}/\text{g}$). Replacing sugar with low-calorie or non-caloric substitutes allows for the formulation of dietetic and weight-control food products. * **Prevention of Dental Caries:** Traditional sugars are readily fermented by oral bacteria (such as *Streptococcus mutans*) into organic acids that erode tooth enamel. Most sugar substitutes are non-cariogenic as they cannot be fermented by oral flora.

Sorbitol

Sorbitol ($\text{C}_6\text{H}_{14}\text{O}_6$) is a polyol, or sugar alcohol, produced industrially by the catalytic hydrogenation of D-glucose, converting the aldehyde group into a secondary alcohol group.

* **Sweetness Profile:** Sorbitol exhibits a sweetness intensity of approximately 50% to 70% (or 0.5 to 0.7 times) relative to sucrose. * **Caloric Value and Metabolism:** It provides approximately $2.6\ \text{kcal}/\text{g}$. Because its absorption in the human small intestine occurs via passive diffusion, it undergoes slow metabolism and leads to minimal glycemic response. * **Uses:** Sorbitol is utilized in sugar-free chewing gums, mints, diabetic chocolates, dietetic baked goods, and toothpaste. Beyond its sweetening ability, it functions as a humectant to maintain product moisture and produces a negative heat of solution, resulting in a cooling sensation in the mouth.

Saccharin

Saccharin (1,1-dioxo-1,2-benzothiazol-3-one, $\text{C}_7\text{H}_5\text{NO}_3\text{S}$) is an artificial non-nutritive sweetener usually prepared as a sodium or calcium salt to enhance aqueous solubility.

* **Sweetness Profile:** Saccharin is an intense sweetener, possessing a sweetness potency approximately 300 to 500 times greater than sucrose. At higher concentrations, it can exhibit a bitter or metallic aftertaste. * **Caloric Value:** It passes through the human digestive tract unabsorbed or unmetabolized, providing $0\ \text{kcal}/\text{g}$. * **Uses:** Saccharin is used in tabletop sweetener packets, diet carbonated drinks, low-calorie jams, jellies, canned fruits, and pharmaceutical syrups. It possesses thermal stability, rendering it suitable for cooking and baking applications without loss of sweetness. * **Safety Debate:** In the 1970s, toxicological studies reported a higher incidence of urinary bladder tumors in male rats fed high doses of saccharin. This led regulatory bodies to restrict its use or mandate warning labels. However, subsequent mechanistic studies demonstrated that

bladder tumor formation in male rats resulted from a species-specific mechanism involving high urinary calcium phosphate precipitation unique to rodent physiology. Extensive epidemiological studies in humans showed no correlation between saccharin consumption and bladder cancer risk. Consequently, global regulatory agencies removed warning mandates and affirmed the safety of saccharin within established Acceptable Daily Intake (ADI) limits.

Cyclamate

Cyclamate (cyclohexanesulfamic acid or its sodium/calcium salts, $C_6H_{13}NO_3S$) is a synthetic, non-nutritive sweetener.

* **Sweetness Profile:** Cyclamate possesses a sweetening power approximately 30 to 50 times greater than sucrose. Its sweetness profile exhibits a slow onset and long duration with a clean taste relative to saccharin. * **Caloric Value:** It yields $0\ \text{kcal}/\text{g}$ as it provides no usable metabolic energy to the human body. * **Uses:** Cyclamate is utilized in low-calorie beverages, tabletop sweeteners, and processed foods. It is often blended with saccharin in a 10:1 ratio (cyclamate to saccharin); this synergistic combination enhances overall sweetness while mutually masking the off-tastes of both individual compounds.

Comparative Analysis of Sugar Substitutes

The physical, chemical, and sensory characteristics of sorbitol, saccharin, and cyclamate relative to sucrose are summarized below:

Property	Sucrose (Reference)	Sorbitol	Saccharin	Cyclamate
Chemical Type	Disaccharide	Sugar alcohol (Polyol)	Synthetic sulfimide	Synthetic sulfamate
Relative Sweetness	$1.0\times$	$0.5-0.7\times$	$300-500\times$	$30-50\times$
Caloric Value	$4.0\ \text{kcal}/\text{g}$	$2.6\ \text{kcal}/\text{g}$	$0\ \text{kcal}/\text{g}$	$0\ \text{kcal}/\text{g}$
Primary Functional Role	Bulk sweetener & energy source	Bulk sweetener, humectant, low-glycemic agent	High-intensity sweetener	High-intensity sweetener
Key Applications	Standard foods and drinks	Diabetic foods, chewing gum, toothpaste	Diet sodas, tabletop packets, baked goods	Synergistic sweetener blends, diet beverages

3. Food Colours, Proteins, Carbohydrates, Lipids and Emulsions

Food Colours

Food colours are natural or synthetic substances added to food items to enhance their visual appeal, replace colour lost during industrial processing and storage, ensure batch-to-batch colour consistency, and provide identity to processed products. Consumers strongly associate visual appearance with food quality, freshness, and flavour; hence, colour additives play a critical role in modern food chemistry and technology.

Food colours are broadly classified into two categories: natural colours and synthetic colours.

Natural Colours

Natural food colours are organic pigments extracted from biological sources such as plants, animals, fungi, and micro-organisms, or produced through thermal treatment of natural food ingredients.

Turmeric, derived from the rhizome of *Curcuma longa*, owes its characteristic bright yellow hue to curcumin. Curcumin is a lipophilic polyphenol structurally defined as diferuloylmethane, possessing symmetrical feruloyl groups linked by a central methylene carbon. It displays keto-enol tautomerism, existing predominantly in the enol form in organic solvents and alkaline aqueous media.

Beetroot red is extracted from the root vegetable *Beta vulgaris*. The principal pigment present is betanin, a water-soluble nitrogenous pigment belonging to the betalain family. Betanin consists of a betanidin aglycone bound to a glucose moiety. It provides a vivid red to purple colouration and is sensitive to heat, oxygen, and pH changes, operating optimally at a pH range between 4.0 and 5.0.

Caramel brown is produced through the controlled thermal degradation of carbohydrates such as sucrose, glucose, or fructose in the presence of acidic, alkaline, or ammonium catalysts. Caramelisation generates a complex mixture of colloidal, high-molecular-weight polymeric pigments alongside volatile aroma compounds. Caramel colours are widely used in beverages, baked goods, and sauces.

Chlorophyll green is the primary photosynthetic pigment extracted from green plant tissues such as spinach and alfalfa. Chlorophyll molecules consist of a substituted tetrapyrrole porphyrin ring coordinated to a central magnesium ion (Mg^{2+}), attached to a long lipophilic phytol chain ($\mathrm{C_{20}H_{39}OH}$). Under acidic conditions or heat treatment, the magnesium ion is easily displaced by two hydrogen ions, forming pheophytin, which exhibits an unappealing olive-brown shade.

Carotene orange refers to lipophilic carotenoid pigments, most notably β-carotene. The chemical structure of β-carotene ($\mathrm{C_{40}H_{56}}$) consists of a long, conjugated polyene chain terminated by two β-ionone rings. The extensive network of conjugated double bonds absorbs light in the visible blue spectrum, resulting in a vibrant yellow-orange appearance. β-Carotene also serves as a nutritional provitamin A precursor.

Synthetic Colours

Synthetic food colours are chemically synthesized organic dyes, predominantly coal-tar derivatives, engineered to exhibit high chemical stability, intense colouring power, uniform shade, and low production costs compared to natural extracts.

Tartrazine (Yellow 5) is a water-soluble synthetic azo dye. Chemically, it is the trisodium salt of 3-carboxy-5-hydroxy-1-(p-sulfophenyl)-4-(p-sulfophenylazo)pyrazole ($\mathrm{C_{16}H_9N_4Na_3O_9S_2}$). The presence of the azo functional group ($-\mathrm{N=N}-$) linked to aromatic rings with sulfonic acid groups provides strong water solubility and stability across varied thermal and pH conditions.

Erythrosine (Red 3) is an organo-iodine synthetic dye belonging to the xanthene class. Structurally, it is the disodium salt of tetraiodofluorescein ($\mathrm{C_{20}H_6I_4Na_2O_5}$). The incorporation of four iodine atoms onto the xanthene nucleus imparts an intense cherry-red colouration, frequently used in canned fruits, confectionery, and baked goods.

Brilliant Blue FCF (Blue 1) is a synthetic triarylmethane dye. Its molecular formula is $\mathrm{C_{37}H_{34}N_2Na_2O_9S_3}$. The structure features three aromatic rings bound to a central carbon atom, stabilized by sulfonate groups that grant high water solubility and physical resistance to light degradation.

Composition and Safety Rules for Synthetic Colours

Synthetic food dyes contain distinct chromophores, such as azo ($-\mathrm{N=N}-$), xanthene, or triarylmethane systems, attached to auxochromes like hydroxyl ($-\mathrm{OH}$) and sulfonate ($-\mathrm{SO_3Na}$) groups. These auxochromes enhance solubility and alter light absorption wavelengths.

Because synthetic dyes are xenobiotic compounds, regulatory bodies such as the Joint FAO/WHO Expert Committee on Food Additives (JECFA) and national food safety authorities enforce strict legal limits on their usage. Regulatory frameworks establish an Acceptable Daily Intake (ADI), expressed in milligrams per kilogram of body weight per day ($\mathrm{mg/kg\ bw/day}$).

Permitted synthetic dyes must satisfy high purity standards, typically requiring a dye content of not less than 85%, with strict upper limits on unreacted intermediates, heavy metals (lead $\leq 10\ \mathrm{ppm}$, arsenic $\leq 3\ \mathrm{ppm}$), and subsidiary colours. In food products, the total concentration of permitted synthetic food colours must not exceed prescribed legal limits, commonly set at $100\ \mathrm{mg/kg}$ ($100\ \mathrm{ppm}$) in final ready-to-consume foods, extending to a maximum of $200\ \mathrm{mg/kg}$ in specified canned or processed products. Unpermitted dyes, such as Metanil Yellow and Rhodamine B, are strictly prohibited due to established hepatotoxic, neurotoxic, and carcinogenic risks. Labeling laws demand explicit declaration of added synthetic colours by their specific names or international E-numbers on consumer packaging.

Proteins and Meat

Proteins in Food

Proteins are high-molecular-weight biopolymers composed of monomeric units called α-amino acids. Each α-amino acid possesses a central carbon atom (α-carbon) covalently bonded to an amino group ($-\mathrm{NH_2}$), a carboxyl group ($-\mathrm{COOH}$), a hydrogen atom ($-\mathrm{H}$), and a distinctive side-chain group ($-\mathrm{R}$).

Individual amino acids are linked covalently through peptide bonds. A peptide bond is an amide linkage formed via a condensation reaction between the α-carboxyl group of one amino acid and the α-amino group of another, with the elimination of a water molecule:

$$\mathrm{R_1–CH(NH_2)–COOH + R_2–CH(NH_2)–COOH \longrightarrow R_1–CH(NH_2)–CO–NH–CH(R_2)–COOH + H_2O}$$

Amino acids present in human nutrition are divided into essential and non-essential categories:

1. Essential amino acids cannot be synthesized endogenously by the human body at a rate sufficient to satisfy metabolic demand and must be obtained directly from dietary sources. These include histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine.
2. Non-essential amino acids can be synthesized within human metabolic pathways from metabolic intermediates. These include alanine, arginine, asparagine, aspartic acid, cysteine, glutamic acid, glutamine, glycine, proline, serine, and tyrosine.

Proteins serve vital physiological and biochemical roles in the human body: - **Structural Function**: Collagen, keratin, and elastin form the structural scaffold of connective tissues, skin, bones, and cartilage. - **Catalytic Function**: Enzymes are globular proteins that act as biological catalysts to accelerate metabolic reactions. - **Transport and Storage**: Hemoglobin transports oxygen through the circulatory system, while ferritin stores iron in liver cells. - **Immunological Defence**: Immunoglobulins (antibodies) recognize and neutralize foreign pathogens. - **Metabolic and Regulatory Function**: Peptide hormones such as insulin regulate cellular glucose homeostasis. Proteins also supply metabolic energy ($4\ \mathrm{kcal/g}$) when carbohydrate and lipid intake is insufficient.

Meat as a Source of Protein

Meat represents the skeletal muscle tissue of slaughtered animals, poultry, and game, serving as a nutrient-dense dietary protein source.

The approximate chemical composition of lean skeletal muscle meat consists of: - Water: $70 - 75\%$ - Protein: $18 - 22\%$ - Lipids (Fat): $2 - 10\%$ - Minerals (Ash): $1.0 - 1.5\%$ - Carbohydrates (Glycogen): $< 1.0\%$

The major structural proteins in meat are categorized into myofibrillar proteins (actin and myosin, responsible for muscle contraction), sarcoplasmic proteins (myoglobin and metabolic enzymes, soluble in water), and stromal or connective tissue proteins (collagen and elastin, providing tissue rigidity).

When meat is subjected to cooking, several distinct physical and chemical transformations occur:

1. Protein Denaturation: Heat alters the secondary, tertiary, and quaternary structures of muscle proteins. Myosin denatures at temperatures between $50^\circ\mathrm{C}$ and $55^\circ\mathrm{C}$, leading to structural contraction, while actin denatures around $66 - 80^\circ\mathrm{C}$. Collagen undergoes thermal hydrolysis at elevated temperatures ($60 - 70^\circ\mathrm{C}$ in the presence of moisture), transforming its insoluble triple-helix fibril structure into soluble gelatin.
2. Loss of Water: Thermally induced shrinkage of the myofibrillar protein gel network squeezes water out of the interfilamental spaces, resulting in exudative cooking loss and reduced drip volume.
3. Browning and Colour Changes: Heat causes the denaturation of myoglobin, converting the native red oxymyoglobin pigment into denatured globin hemichrome, which imparts a characteristic brown-grey colour to cooked meat. Simultaneously, non-enzymatic Maillard reactions occur between amino groups of meat proteins and traces of reducing sugars, yielding complex brown pigments and savoury volatile aromas.

The nutritive value of meat is exceptionally high. Meat provides a complete protein source containing all essential amino acids in proportions optimal for human tissue synthesis, exhibiting high biological value and digestibility. Additionally, meat delivers highly bioavailable heme iron, zinc, selenium, phosphorus, and essential B-complex vitamins, notably cyanocobalamin (Vitamin B_{12}).

Carbohydrates

Carbohydrates are organic polyhydroxy aldehydes or polyhydroxy ketones, or substances that yield such compounds upon hydrolysis. They are categorized based on structural complexity into simple sugars (monosaccharides and disaccharides), oligosaccharides, and structural or storage polysaccharides (starch and dietary fibre).

Structures and Classes of Carbohydrates

Monosaccharides are simple single-unit sugars with the general formula $(\mathrm{CH_2O})_n$. D-Glucose (an aldohexose) and D-fructose (a ketohexose) are common examples. Disaccharides consist of two monosaccharide units joined by a glycosidic bond:

1. Sucrose consists of α-D-glucopyranose and β-D-fructofuranose linked by an α, β-(1$\rightarrow$2) glycosidic bond.
2. Lactose consists of β-D-galactopyranose linked to α-D-glucopyranose via a β-(1$\rightarrow$4) glycosidic bond.
3. Maltose consists of two α-D-glucopyranose units linked via an α-(1$\rightarrow$4) glycosidic bond.

Starch is the primary storage carbohydrate in plants, existing as a mixture of two distinct homopolymers of α-D-glucose:

- Amylose is a predominantly linear polymer consisting of D-glucose units linked exclusively by α-(1$\rightarrow$4) glycosidic bonds, forming a helical macromolecular structure. - Amylopectin is a highly branched polymer containing linear chains of D-glucose linked by α-(1$\rightarrow$4) glycosidic bonds, with branch points occurring via α-(1$\rightarrow$6) glycosidic bonds every 24 to 30 glucose residues.

Dietary fibre encompasses plant carbohydrate polymers resistant to digestion and absorption in the human small intestine. The primary structural component of plant cell walls is cellulose. Cellulose is a linear homopolymer of D-glucose units linked by β-(1$\rightarrow$4) glycosidic bonds:

$$-[\,\mathrm{C_6H_{10}O_5}\,]_n- \text{ linked via } \beta\text{-}(1 \rightarrow 4) \text{ glycosidic bonds}$$

The β-configuration forms linear, extended chains stabilized by inter- and intra-molecular hydrogen bonding. Human digestive enzymes (such as α-amylase) are stereospecific for α-glycosidic linkages and cannot hydrolyze the β-(1$\rightarrow$4) linkages of cellulose. Other dietary fibres include hemicelluloses, pectins (polygalacturonic acid polymers), and plant gums.

Roles of Carbohydrates in Food

Carbohydrates serve three fundamental functions in food systems:

- **Energy Provision**: Digestible carbohydrates (sugars and starch) yield $4\ \mathrm{kcal/g}$ ($17\ \mathrm{kJ/g}$) upon cellular oxidation, serving as the primary metabolic fuel for human physiological processes. - **Texture and Structure**: Starch granules undergo gelatinization when heated in water, absorbing liquid and swelling to form viscous gels used as thickeners, stabilizers, and binding agents in soups, sauces, and puddings. Dietary fibre provides physical bulk, structural firmness to fruits and vegetables, and water retention capacity in processed foods. - **Sweetness**: Mono- and disaccharides impart sweetness to food products. The relative sweetness intensity varies among sugars: taking sucrose as a baseline standard with a value of 100, fructose exhibits a relative sweetness of approximately 140–170, glucose ranks at 70–80, and lactose exhibits a low relative sweetness of roughly 15–20.

Lipids and Emulsions

Lipids in Food

Lipids are a chemically heterogeneous group of nonpolar organic compounds that are insoluble in water but soluble in nonpolar organic solvents such as hexane, ether, and chloroform.

Fats and oils are simple lipids chemically defined as triacylglycerols (triglycerides). Triacylglycerols are triesters composed of a single glycerol backbone molecule esterified with three fatty acid chains:

$$\begin{aligned} \mathrm{CH_2–OH} \quad &+ \quad \mathrm{HOOC–R_1} \quad \longrightarrow \quad \mathrm{CH_2–O–CO–R_1} \\ \mathrm{CH_2–OH} \quad &+ \quad \mathrm{HOOC–R_2} \quad \longrightarrow \quad \mathrm{CH_2–O–CO–R_2} \quad + \quad 3\,\mathrm{H_2O} \\ \mathrm{CH_2–OH} \quad &+ \quad \mathrm{HOOC–R_3} \quad \longrightarrow \quad \mathrm{CH_2–O–CO–R_3} \end{aligned}$$

Fats are solid or semi-solid at room temperature ($20^\circ\mathrm{C}$) due to a high content of saturated fatty acids, whereas oils are liquids owing to a high concentration of unsaturated fatty acids.

Fatty acids are classified based on the presence and degree of double bonds within their hydrocarbon chains:

Saturated fatty acids contain no carbon-carbon double bonds ($\mathrm{C=C}$) in their hydrocarbon chain. They have the general molecular formula $\mathrm{C_nH_{2n+1}COOH}$. Common examples include palmitic acid ($\mathrm{C_{15}H_{31}COOH}$) and stearic acid ($\mathrm{C_{17}H_{35}COOH}$). Their linear hydrocarbon chains pack tightly together, maximizing van der Waals interactions and resulting in higher melting points.

Unsaturated fatty acids contain one or more carbon-carbon double bonds in their hydrocarbon chain. Monounsaturated fatty acids (MUFAs) possess a single double bond (e.g., oleic acid, $\mathrm{C_{17}H_{33}COOH}$), while polyunsaturated fatty acids (PUFAs) possess two or more double bonds (e.g., linoleic acid, $\mathrm{C_{17}H_{31}COOH}$). Naturally occurring unsaturated fatty acids generally exist in the *cis*-configuration, which introduces a structural bend or "kink" in the carbon chain. This kink hinders close molecular packing, lowering the melting point.

Trans-fats are unsaturated fatty acids that contain one or more non-conjugated double bonds in the *trans*-geometric configuration. Trans-fats are generated industrially during the partial hydrogenation of unsaturated vegetable oils, a process designed to convert liquid oils into semi-solid fats for commercial baking and frying. The elevated temperatures and metal catalysts (such as nickel) used during partial hydrogenation cause geometric isomerization of naturally occurring *cis*-double bonds into thermodynamically stable *trans*-configurations:

$$\mathrm{–CH_2–CH{=}CH–CH_2–\ (cis)} \xrightarrow{\text{catalytic hydrogenation / heat}} \mathrm{–CH_2–CH{=}CH–CH_2–\ (trans)}$$

Trans-fats are considered harmful to human health because their linear *trans*-conformation mimics saturated fatty acids. Clinical and epidemiological studies show that consumption of trans-fats elevates circulating levels of low-density lipoprotein (LDL) cholesterol while simultaneously decreasing high-density lipoprotein (HDL) cholesterol. This combined effect causes vascular endothelial dysfunction, accelerates atherogenesis, and significantly increases the risk of coronary heart disease and metabolic syndrome.

Compound lipids are esters of fatty acids containing non-lipid structural groups attached to an alcohol moiety:

- Phospholipids contain a glycerol molecule esterified to two fatty acids and a phosphate group linked to an amino alcohol derivative. A prime example is phosphatidylcholine (lecithin). - Glycolipids consist of fatty acids bound to a carbohydrate moiety and a sphingosine or glycerol backbone. - Lipoproteins are macromolecular complexes of lipids and structural proteins (apolipoproteins) that transport hydrophobic lipids through blood plasma.

Derived lipids are substances produced via the enzymatic or chemical hydrolysis of simple and compound lipids, retaining lipid-like solubility characteristics. Examples include free fatty acids, glycerol, fat-soluble vitamins (Vitamins A, D, E, K), and sterols such as cholesterol.

Emulsions

An emulsion is a thermodynamically unstable colloidal system consisting of two immiscible liquids, where one liquid (the dispersed phase) is suspended as microscopic droplets throughout the second liquid (the continuous phase).

Food emulsions are classified into two principal geometric structural types:

- Oil-in-Water (O/W) Emulsion: Nonpolar oil droplets are dispersed throughout a continuous aqueous phase. Examples include milk, cream, mayonnaise, and salad dressings. - Water-in-Oil (W/O) Emulsion: Water droplets are dispersed throughout a continuous nonpolar oil phase. Examples include butter and margarine.

Emulsifying agents (emulsifiers) are surface-active amphiphilic molecules containing both a hydrophilic (water-attracting polar) domain and a lipophilic (oil-attracting nonpolar) domain within their chemical structure.

Lecithin (phosphatidylcholine) is a classic food emulsifier. Structurally, lecithin features a hydrophilic zwitterionic polar head group (consisting of a negatively charged phosphate group bonded to a positively charged choline group) and two lipophilic nonpolar hydrocarbon tails formed by fatty acid chains.

When added to a mixture of oil and water, lecithin molecules spontaneously align themselves at the oil-water interface. The nonpolar fatty acid tails dissolve in the hydrophobic oil phase, while the polar hydrophilic head group extends into the aqueous phase:

$$\text{Interfacial tension reduction: } \gamma_{\text{oil-water}} \longrightarrow \gamma_{\text{emulsion}} \ll \gamma_{\text{oil-water}}$$

This interfacial orientation lowers the interfacial free energy and dynamic interfacial tension (γ) between the two phases. Furthermore, emulsifiers form a mechanical, steric, or electrostatic protective barrier around the dispersed droplets. This protective film prevents droplet collisions from leading to coalescence, flocculation, creaming, or phase separation, thereby preserving the physical stability of the food emulsion over time.

4. Browning Reactions

Browning reactions refer to complex chemical and biochemical processes that cause food items to develop brown pigments during cutting, post-harvest handling, thermal cooking, or extended storage. Browning significantly alters the visual appearance, organoleptic attributes, flavour profile, texture, and nutritional quality of fresh and processed foods.

Browning reactions are classified into two broad mechanistic pathways: enzymatic browning and non-enzymatic browning.

Enzymatic Browning

Enzymatic browning occurs primarily in fresh fruits, vegetables, and tubers (such as apples, bananas, pears, avocadoes, and potatoes) when plant tissues are mechanically damaged by cutting, peeling, bruising, or crushing.

Mechanism of Enzymatic Browning

The biochemical process is mediated by the intracellular enzyme polyphenol oxidase (PPO), also known as tyrosinase, catecholase, or monophenol monooxygenase. PPO is a copper-containing metalloenzyme localized within cellular plastids (chloroplasts and leucoplasts), whereas phenolic substrates are stored separately inside vacuoles.

Mechanical damage disrupts cellular compartmentalization, bringing PPO into direct contact with endogenous phenolic substrates and atmospheric oxygen (O_2). The reaction proceeds via a two-stage oxidation pathway:

First, PPO acts via its cresolase activity to catalyze the hydroxylation of monophenols to form *ortho*-diphenols (o-diphenols):

$$\text{Monophenol} + O_2 + AH_2 \xrightarrow{\text{Polyphenol Oxidase (Cresolase)}} \text{ortho-Diphenol} + A + H_2O$$

Second, PPO acts via its catecholase activity to oxidize the *ortho*-diphenols into reactive *ortho*-quinones (o-quinones):

$$2\ \text{ortho-Diphenol} + \text{O}_2 \xrightarrow{\text{Polyphenol Oxidase (Catecholase)}} 2\ \text{ortho-Quinone} + 2\ \text{H}_2\text{O}$$

The resulting *ortho*-quinones are highly electrophilic, reactive intermediates. In subsequent non-enzymatic steps, quinones spontaneously undergo irreversible condensation and polymerization reactions with other quinones, amino acids, and proteins, forming high-molecular-weight, insoluble brown nitrogenous pigments termed melanins.

Prevention of Enzymatic Browning

Controlling enzymatic browning is essential for preserving the commercial quality of fresh produce. Prevention methods focus on inactivating the PPO enzyme, eliminating molecular oxygen, or adjusting chemical conditions:

- **Blanching**: Heat treatment (such as steaming or boiling water immersion for short durations) thermal denatures the tertiary structure of the PPO protein, rendering the copper metalloenzyme biologically inactive. - **Addition of Ascorbic Acid or Lemon Juice**: Adding lemon juice lowers the pH of the system below 4.0. PPO exhibits optimal catalytic activity between $\text{pH}\ 6.0$ and 7.0; reducing pH significantly decreases enzyme kinetics. Furthermore, ascorbic acid (Vitamin C) acts as a potent reducing agent, chemically reducing reactive *ortho*-quinones back into colorless *ortho*-diphenols before polymerizing into melanins:

$$\text{ortho-Quinone} + \text{Ascorbic Acid} \longrightarrow \text{ortho-Diphenol} + \text{Dehydroascorbic Acid}$$

- **Exclusion of Air**: Submerging freshly cut fruits and vegetables in water, applying sugar syrups, vacuum packaging, or flushing packaging with an inert gas like nitrogen (N_2) deprives PPO of molecular oxygen, an essential co-substrate for oxidation.

Non-Enzymatic Browning

Non-enzymatic browning occurs without catalytic enzymes and is primarily driven by thermal energy during cooking, baking, roasting, and frying, or during prolonged storage. The two main non-enzymatic pathways are the Maillard reaction and caramelisation.

The Maillard Reaction

The Maillard reaction is a complex cascade of chemical reactions initiated by the nucleophilic addition of an unprotonated amino group (from amino acids, peptides, or proteins) to the electrophilic carbonyl group of a reducing sugar (such as glucose, fructose, or ribose).

The reaction pathway proceeds through three main stages:

Initial Stage

The unprotonated amino group (R–NH_2) reacts with the open-chain carbonyl group of a reducing sugar to yield an unstable addition compound, an N-substituted glycosylamine, with the loss of a water molecule:

$$\text{Sugar–C=O} + \text{R–NH}_2 \rightleftharpoons \text{Glycosylamine} + \text{H}_2\text{O}$$

The resulting N-glycosylamine undergoes an irreversible acid-base catalyzed isomerization: - If an aldose sugar is used, it undergoes the Amadori Rearrangement to form a 1-amino-1-deoxy-2-ketose. - If a ketose sugar is used, it undergoes the Heyns Rearrangement to form a 2-amino-2-deoxy-aldose.

Intermediate Stage

Amadori and Heyns products undergo further transformations depending on pH and temperature: - At neutral or acidic pH, 1,2-enolisation yields 3-deoxyosones, which dehydrate into heterocyclic intermediates such as 5-hydroxymethylfurfural (HMF) or furfural. - At alkaline pH, 2,3-enolisation forms 1-deoxyosones, which fragment into volatile dicarbonyl compounds such as diacetyl, acetol, and pyruvaldehyde.

Concurrently, Strecker Degradation occurs: α-dicarbonyl intermediates react with α-amino acids to cause oxidative decarboxylation, generating carbon dioxide (CO_2), Strecker aldehydes, and α-aminoketones. Strecker aldehydes contribute directly to characteristic cooked aromas.

Final Stage

Unstable dicarbonyls, aminoketones, and furfural derivatives undergo aldol condensations and polymerization with cellular nitrogen compounds, ultimately forming high-molecular-weight, insoluble brown nitrogenous polymers termed melanoidins.

The Maillard reaction generates key visual pigments and aroma-active volatile compounds (such as pyrazines, pyrroles, oxazoles, and thiazoles), giving baked bread crusts, roasted coffee beans, chocolate, and seared meats their characteristic sensory appeal.

Caramelisation

Caramelisation is the thermal degradation of sugars occurring in the absence of amino compounds. It requires high temperatures, typically above the melting point of the sugar (above $160^\circ C$ for sucrose, $110^\circ C$ for fructose, and $160^\circ C$ for glucose).

The process involves thermal elimination reactions, enolisation, isomerization, oxidation, and fragmentation:

1. Heating induces sugar melting followed by mutarotation and enolisation.
2. Dehydration reactions eliminate water molecules, generating internal anhydro-sugar rings and dicarbonyl species.
3. Intramolecular condensation and cross-linking yield three major distinct polymeric brown fraction classes:

- Caramelan ($C_{12}H_{18}O_9$): A lower-molecular-weight, light brown, water-soluble fraction formed after the loss of two water molecules from sucrose. - Caramelen ($C_{36}H_{50}O_{25}$): A larger pigment polymer formed after additional heating and dehydration. - Caramelin ($C_{125}H_{188}O_{80}$): A high-molecular-weight, dark brown, insoluble pigment polymer produced by extensive thermal processing.

Caramelisation also yields pleasant volatile aroma compounds, including hydroxyfuranones, diacetyl, and maltol, which impart sweet, buttery, and toasted sensory characteristics to confectionery and roasted food products.

Chemical and Biochemical Influence of Browning Reactions

Browning reactions alter the physicochemical and organoleptic properties of fresh and processed foods:

Influence on Colour

Enzymatic browning often causes undesirable darkening in cut fresh produce, leading to economic losses. Conversely, non-enzymatic Maillard and caramelisation reactions create desirable golden-brown to deep-brown shades in bread crusts, potato chips, biscuits, roasted coffee, and dark beers.

Influence on Flavour

Enzymatic browning produces astringent off-flavours caused by quinone accumulation. In contrast, controlled non-enzymatic browning generates desirable aroma profiles: - Maillard reactions yield savory, nutty, and roasted notes due to pyrazines, pyridines, and Strecker aldehydes. - Caramelisation yields sweet, buttery, and burnt-sugar notes from furans and diacetyl.

However, excessive thermal processing causes over-browning, imparting acrid, bitter, and burnt off-flavours.

Influence on Texture

Non-enzymatic browning alters textural matrix properties. During baking, Maillard cross-linking between proteins and carbohydrates, combined with thermal moisture evaporation, forms a dry, crisp outer crust on baked goods. In meat processing, thermal protein denaturation and cross-linking influence juice retention and final tenderness.

Nutritional and Biochemical Consequences

- **Loss of Essential Amino Acids**: The Maillard reaction preferentially involves the ϵ-amino group of free or protein-bound lysine residues. Because the human body cannot utilize Amadori-bound lysine, this reaction reduces the bioavailability of essential amino acids and lowers the overall protein digestibility of thermally processed foods. - **Vitamin Destruction**: Thermal processing and enzymatic oxidation deplete ascorbic acid (Vitamin C), reducing overall micronutrient content. - **Formation of Toxic Compounds**: High-temperature Maillard reactions between reducing sugars and free asparagine (particularly in fried potato products and baked cereals) generate acrylamide ($\mathrm{CH_2{=}CH{-}CO{-}NH_2}$), a classified neurotoxin and potential human carcinogen. - **Antioxidant Activity**: Conversely, early- and intermediate-stage Maillard reaction products and melanoidins exhibit free-radical scavenging capabilities and metal-chelating properties, providing antioxidant benefits in processed food matrices.

5. Preservation of Foods

Why Food Needs Preservation

Food items are complex biological matrices composed of proteins, carbohydrates, lipids, water, vitamins, and minerals. From the moment of harvesting, slaughtering, or processing, raw food materials undergo progressive deterioration. The primary causes of food spoilage can be categorized into three major chemical and biological factors:

1. **Micro-organisms**: Bacteria, moulds, and yeasts metabolize nutrients present in the food matrix. Microbial growth leads to the production of unpalatable off-flavours, foul odours, gas formation, structural breakdown, and toxic secondary metabolites such as mycotoxins or bacterial exotoxins, rendering the food unsafe for consumption.
2. **Enzymes**: Endogenous enzymes naturally present within plant and animal tissues continue their catalytic actions post-harvest or post-mortem. Oxidases, lipases, proteases, and carbohydrases catalyze the enzymatic breakdown of structural components, cellular autolysis, loss of firmness, and unwanted color transformations.
3. **Chemical Changes**: Non-enzymatic chemical reactions occur spontaneously upon exposure of food components to atmospheric oxygen, light, and heat. A prominent example is lipid autoxidation, where unsaturated fatty acids react with atmospheric oxygen to form unstable hydroperoxides that decompose into volatile aldehydes and ketones, causing rancidity and loss of nutritional value.

General Principles of Food Preservation

The physical and chemical preservation of food relies on establishing conditions that either completely eliminate or substantially retard microbial proliferation, enzyme activities, and spontaneous chemical degradation. The foundational principles governing food preservation include:

- **Removing Water**: Water acts as a solvent and reactant in biochemical processes. Controlling the water activity (a_w) of a food matrix limits the unbound water available for microbial cellular functions and enzymatic catalysis. - **Lowering Temperature**: Decreasing the thermal energy of the food system decreases the kinetic energy of molecules, thereby lowering the reaction rate constant k according to the Arrhenius relationship:

$$k = Ae^{-\frac{E_a}{RT}}$$

where A is the pre-exponential factor, E_a is the activation energy, R is the universal gas constant, and T is the absolute temperature. Reducing T reduces microbial metabolic activity and enzymatic rates. - **Excluding Air**: Eliminating gaseous oxygen (O_2) suppresses the metabolic pathways of obligate aerobic micro-organisms and prevents free-radical chain reactions associated with lipid autoxidation. - **Adding Preservatives**: Introducing specific chemical agents inhibits microbial growth by disrupting cellular membrane integrity, altering intracellular pH, or inactivating key metabolic enzymes. Antioxidants are incorporated to scavenge free radicals and prevent chemical oxidation. - **Destroying Micro-organisms**: Applying high thermal

energy or ionizing radiation denatures essential microbial proteins, disrupts nucleic acid structures, and inactivates cellular enzymes, resulting in complete sterility or a significant reduction in microbial load.

Physical Methods of Food Preservation

Refrigeration and Freezing

Refrigeration involves maintaining food at temperatures between 0°C and 5°C. This low-temperature environment depresses the rates of enzymatic reactions and slows microbial cell division without altering the physical state of water in the food.

Freezing lowers the temperature to -18°C or below. At these temperatures, liquid water undergoes a phase transition to solid ice, which drastically reduces the water activity a_w. Because liquid water is required for microbial transport mechanisms, microbial growth is completely arrested.

Examples of preserved foods: Fresh liquid milk, fresh fruits, and leafy vegetables are preserved via refrigeration. Raw fish fillets, poultry cuts, green peas, and ready-to-eat meals are preserved by freezing.

Drying and Dehydration

Drying and dehydration involve removing water from the food matrix through evaporation or sublimation. Reducing the moisture content lowers the water activity a_w below 0.60, a threshold below which most bacteria, yeasts, and moulds cannot grow or reproduce.

Examples of preserved foods: Sun-dried raisins and figs, spray-dried milk powder, dehydrated instant coffee, and dried beef jerky.

Heating: Pasteurisation and Sterilisation

Thermal processing uses heat energy to denature proteins and permanently inactivate metabolic enzymes in micro-organisms.

Pasteurisation is a mild heat treatment designed to eliminate pathogenic vegetative micro-organisms and reduce spoilage organisms without drastically affecting the sensory quality of the food. For instance, High-Temperature Short-Time (HTST) pasteurisation typically exposes the liquid food to 72°C for 15 seconds.

Sterilisation is a high-intensity thermal process, typically exceeding 100°C (often 121°C under pressure), that destroys all viable vegetative micro-organisms and highly resistant bacterial endospores (such as those of *Clostridium botulinum*). Ultra-High Temperature (UHT) processing exposes products to 135°C to 150°C for 2 to 5 seconds.

Examples of preserved foods: Pasteurised fluid milk, beer, and fruit juices. Sterilised UHT milk and liquid cream packaged ambiently.

Canning

Canning combines thermal sterilisation with hermetic sealing. The food product is placed in a clean container, exhausted to remove trapped air, sealed airtight, and subjected to retort heating under steam pressure to achieve commercial sterility. The hermetic seal prevents re-contamination by airborne micro-organisms and excludes atmospheric oxygen.

Examples of preserved foods: Canned green beans, canned tuna, canned sweetcorn, and canned soups.

Irradiation

Food irradiation involves exposing the food product to controlled levels of ionizing radiation, such as gamma rays emitted from radioisotopes (^{60}Co or ^{137}Cs), X-rays, or high-energy electron beams. Ionizing radiation generates reactive radiolytic species that break phosphodiester bonds in microbial DNA, preventing cellular replication and killing spoilage organisms and pests without causing a significant rise in product temperature.

Examples of preserved foods: Irradiated dried spices, irradiated potatoes (to inhibit sprouting), and irradiated fresh strawberries (to prevent mould decay).

Packaging: Vacuum and Modified Atmosphere Packaging

Vacuum packaging involves placing the food item inside a low-gas-permeability polymer film, evacuating atmospheric air from the package, and heat-sealing the enclosure. The removal of O_2 suppresses aerobic spoilage bacteria and prevents oxidative rancidity.

Modified Atmosphere Packaging (MAP) alters the gaseous composition inside the sealed package. Atmospheric air is flushed out and replaced by a tailored mixture of gases, typically consisting of nitrogen (N_2), carbon dioxide (CO_2), and trace oxygen (O_2). CO_2 dissolves into the aqueous phase of the food to form carbonic acid, which lowers the surface pH and inhibits microbial cell division, while N_2 acts as an inert filler gas to prevent package collapse.

Examples of preserved foods: Vacuum-packed cured bacon and hard cheeses; MAP-packaged fresh salad mixes, raw meat cuts, and fried potato chips.

Chemical Methods of Food Preservation

Adding Salt and Sugar

High concentrations of sodium chloride (NaCl) or sucrose added to a food system raise the extracellular solute concentration, creating a hypertonic medium relative to the microbial cytoplasm. The osmotic pressure Π exerted by the solute concentration C is described by the van 't Hoff equation:

$$\Pi = iCRT$$

where i is the van 't Hoff factor of the solute, R is the gas constant, and T is the temperature.

Because the osmotic pressure outside the cell exceeds the internal pressure, water molecules diffuse out of the microbial cell across the semipermeable cell membrane toward the higher solute concentration. The loss of cellular water leads to plasmolysis, shrinking the cytoplasm, inactivating cellular functions, and preventing cell division.

Examples of preserved foods: Heavily salted codfish and cured ham (using salt); fruit jams, jellies, marmalades, and candied fruits (using high sucrose concentrations of 65% or higher).

Pickling in Vinegar

Pickling relies on lowering the pH of the food matrix by adding organic acids, primarily acetic acid ($\mathrm{CH_3COOH}$) in the form of vinegar. The low pH environment creates an acidic medium ($\mathrm{pH} < 4.6$) that suppresses the growth of spoilage micro-organisms and pathogenic spore-formers. Lipophilic undissociated acetic acid molecules readily diffuse across the microbial cell membrane into the neutral cytoplasm:

$$\mathrm{CH_3COOH} \rightleftharpoons \mathrm{CH_3COO^-} + \mathrm{H^+}$$

Once inside the cytoplasm, the acid dissociates, lowering the internal cytoplasmic pH, disrupting the proton motive force across the membrane, and inactivating critical metabolic enzymes.

Examples of preserved foods: Pickled cucumbers, pickled onions, pickled beetroots, and mixed vegetable pickles.

Chemical Preservatives

Sodium Benzoate

Sodium benzoate ($\mathrm{C_6H_5COONa}$) is the sodium salt of benzoic acid. In an acidic aqueous food system, the benzoate ion establishes an equilibrium with hydronium ions to yield undissociated benzoic acid:

$$\mathrm{C_6H_5COO^-} + \mathrm{H^+} \rightleftharpoons \mathrm{C_6H_5COOH}$$

The acid dissociation constant K_a for benzoic acid is expressed as:

$$K_a = \frac{[\mathrm{C_6H_5COO^-}][\mathrm{H^+}]}{[\mathrm{C_6H_5COOH}]}$$

Rearranging this relationship yields the ratio of undissociated benzoic acid to benzoate ion:

$$\frac{[\mathrm{C_6H_5COOH}]}{[\mathrm{C_6H_5COO^-}]} = \frac{[\mathrm{H^+}]}{K_a}$$

This relation demonstrates that as the hydrogen ion concentration $[\mathrm{H^+}]$ increases (lower pH), the concentration of non-polar, undissociated benzoic acid increases. Undissociated benzoic acid readily dissolves in the lipid bilayer of fungal and bacterial cell membranes, entering the cell to inhibit key respiratory enzymes and anaerobic fermentation pathways (such as glycolysis).

Examples of preserved foods: Carbonated soft drinks, acidic fruit juices, jams, pickles, and salad dressings.

Sorbic Acid

Sorbic acid ($\mathrm{CH_3CH = CHCH = CHCOOH}$) and its soluble salt, potassium sorbate, act as antimicrobial agents against moulds, yeasts, and select bacteria. Like benzoic acid, the active species is the undissociated monocarboxylic acid form. Sorbic acid inhibits cellular enzyme systems by interfering with sulfhydryl groups of key metabolic enzymes, such as dehydrogenases involved in the fatty acid oxidation cycle.

Examples of preserved foods: Processed cheese slices, bakery products (cakes and tortillas), dried fruits, and wines.

Sulphur Dioxide

Sulphur dioxide ($\mathrm{SO_2}$) and its corresponding salts (such as sodium metabisulfite, $\mathrm{Na_2S_2O_5}$) act as dual-function preservatives, operating as antimicrobial agents and reducing agents (antioxidants). When dissolved in an aqueous food phase, sulphur dioxide undergoes a series of pH-dependent equilibria.

First, gaseous sulphur dioxide hydrates to form aqueous sulfurous acid:

$$\mathrm{SO_2 + H_2O \rightleftharpoons H_2SO_3}$$

Sulfurous acid undergoes its first ionization to yield the bisulfite ion:

$$\mathrm{H_2SO_3 \rightleftharpoons HSO_3^- + H^+}$$

The bisulfite ion undergoes a second ionization step to form the sulfite ion:

$$\mathrm{HSO_3^- \rightleftharpoons SO_3^{2-} + H^+}$$

The molecular, un-ionized $\mathrm{SO_2}$ species readily penetrates microbial cell walls and destroys disulfide linkages in vital cellular proteins. Additionally, bisulfite ions ($\mathrm{HSO_3^-}$) react with carbonyl groups of reducing sugars and intermediate quinones, preventing non-enzymatic and polyphenol oxidase-catalyzed enzymatic browning.

Examples of preserved foods: Dried apricots, white wines, dehydrated potato flakes, and fruit juice concentrates.

Antioxidants

Antioxidants are chemical compounds added to lipid-containing foods to prevent oxidative deterioration caused by atmospheric oxygen. Lipid autoxidation occurs via a free-radical chain mechanism initiated by light, heat, or metal ions:

$$\mathrm{RH \xrightarrow{Initiation} R^\bullet + H^\bullet}$$

In the propagation phase, the alkyl radical reacts rapidly with molecular oxygen:

$$\mathrm{R^\bullet + O_2 \xrightarrow{Propagation} ROO^\bullet}$$

The peroxyl radical then abstracts a hydrogen atom from another unsaturated lipid molecule:

$$\mathrm{ROO^{\bullet} + RH \xrightarrow{Propagation} ROOH + R^{\bullet}}$$

Antioxidants (AH) act as primary radical scavengers by donating a hydrogen atom to the peroxyl radical:

$$\mathrm{ROO^{\bullet} + AH \rightarrow ROOH + A^{\bullet}}$$

The resulting antioxidant radical $\mathrm{A^{\bullet}}$ is resonance-stabilized and unreactive, terminating the radical chain reaction:

$$\mathrm{ROO^{\bullet} + A^{\bullet} \rightarrow ROO\text{-}A}$$

Synthetic antioxidants include Butylated Hydroxyanisole (BHA), Butylated Hydroxytoluene (BHT), and Propyl Gallate (PG). Natural antioxidants include α-tocopherol (Vitamin E) and ascorbic acid (Vitamin C).

Examples of preserved foods: Vegetable cooking oils, lard, fried potato snacks, margarine, and breakfast cereals.

Questions

Short form first (2–3 marks), then long form (5–10 marks). The [AU] questions below are from **CHMHCC-501T Biomolecules, 5th Sem, 2023** — topic matches for amino acids, vitamins and lipids only, **not** DSM-401 questions. **No AU question was found on food preservation, sugar substitutes, food colours or browning reactions**; those questions are syllabus-derived, marked [standard].

Short-form questions

S1. [AU] What is meant by essential and non essential amino acids? Give one example of each. — CHMHCC-501T Biomolecules, 5th Sem, 2023, CBCS, Sec A, Unit II, 2 marks.

Answer in brief: Section 1 — essential amino acids cannot be made by the body and must come from food (for example lysine); non-essential amino acids can be made by the body (for example glycine).

S2. [AU] What is isoelectric point of amino acid? What role does it play in electrophoresis? — CHMHCC-501T Biomolecules, 5th Sem, 2023, CBCS, Sec A, Unit II, 2 marks.

Answer in brief: Section 1 — the isoelectric point is the pH at which an amino acid carries no net charge; at this pH it does not move in an electric field, so amino acids can be separated by electrophoresis.

S3. [AU] What are trans-fats? Why are they considered harmful? — CHMHCC-501T Biomolecules, 5th Sem, 2023, CBCS, Sec A, Unit IV, 2 marks.

Answer in brief: Section 3 — trans-fats are unsaturated fats with trans double bonds formed during partial hydrogenation; they raise bad cholesterol and lower good cholesterol, increasing heart disease risk.

S4. [AU] Describe the role of vitamin C in human body. — CHMHCC-501T Biomolecules, 5th Sem, 2023, CBCS, Sec A, Unit V, 2 marks.

Answer in brief: Section 1 — vitamin C is needed for collagen formation, wound healing, iron absorption and as an antioxidant; its deficiency causes scurvy.

Long-form questions

L1. [AU] What are compound and derived lipids? Give examples. — CHMHCC-501T Biomolecules, 5th Sem, 2023, CBCS, Sec A, Unit IV, 2 marks.

What to write: Section 3 — compound lipids contain a non-lipid group as well as fatty acids (for example phospholipids, glycolipids); derived lipids are obtained by hydrolysis of simple or compound lipids (for example fatty acids, sterols, fat-soluble vitamins).

L2. [AU] What are lipids? What are their primary functions in living organism? — CHMHCC-501T Biomolecules, 5th Sem, 2023, CBCS, Sec B, Unit IV, 2 marks.

What to write: Section 3 — lipids are water-insoluble organic compounds (fats, oils, phospholipids, sterols); their functions are energy storage, cell membrane structure, insulation, protection of organs and carrying fat-soluble vitamins.

L3. [standard] What are food additives? Describe vitamins, amino acids and minerals as food additives.

What to write: Section 1 in full — what food additives are, fat-soluble and water-soluble vitamins, essential and non-essential amino acids, and minerals such as calcium, iron, iodine and zinc.

L4. [standard] Describe aroma compounds as food flavours, and sugar substitutes sorbitol, saccharin and cyclamate.

What to write: Section 2 in full — classes of aroma compounds with examples, natural and artificial flavours, and the three sugar substitutes with a comparison.

L5. [standard] Describe food colours, their types and compositions.

What to write: Section 3 (food colours part) — natural colours (turmeric, beetroot, caramel, chlorophyll, carotene) and synthetic colours (tartrazine, erythrosine, brilliant blue), with safety limits.

L6. [standard] Describe proteins and meat, carbohydrates, and lipids and emulsions, and explain browning reactions and their chemical and biochemical influence on colour, flavour and texture in fresh and processed foods.

What to write: Sections 3–4 in full — proteins and meat, carbohydrates, lipids and emulsions, and enzymatic and non-enzymatic browning (Maillard reaction, caramelisation).

L7. [standard] Describe preservation of foods: general principles, physical and chemical methods.

What to write: Section 5 (preservation part) in full — why food spoils, the general principles, physical methods (refrigeration, drying, heating, canning, irradiation, packaging) and chemical methods (salt, sugar, pickling, preservatives, antioxidants).

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.